

# Magda: Information Extraction from German Supermarket Flyers

*Project Proposal — Information Extraction (SoSe 2026)*

<https://github.com/noahsa16/magda>

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## Project Goal

The goal of this project is to build a learning-based system that extracts structured offer data from German supermarket flyers. These flyers are visually dense and noisy, which makes reliable extraction a non-trivial information-extraction problem.

Currently we have a working prototype that behaves as a black box: a flyer PDF is sent in, and structured offer data comes back. Internally this is handled by a Large Language Model (the Gemini API). While this works, it is not a learning-based contribution nothing is trained, and it depends on an external paid API. The aim of this project is therefore to replace the black box with a model we train ourselves, and to use the LLM instead as a tool to automatically label our training data.

The project focuses on:

- Token-level sequence labeling on noisy, visually structured documents.
- Using a layout-aware model that takes the position of words on the page into account, not just the text.
- Using an LLM to automatically generate training labels, rather than as the final extractor.
- Comparing the trained model against the existing LLM black box.

## Entity Schema

We will define our own set of entity types tailored to retail offers, covering roughly the product, its brand, the price, quantity/weight information, discounts, and validity dates, encoded in the BIO format. The exact set of labels will be finalized during the data phase.

## Data Source & Pipeline

Data availability is not a problem: the flyers are already scraped automatically, so we can collect a large number of offers without manual effort. The main challenge is turning the raw flyer images into usable token-level training data. Our pipeline will aim to separate the geometry of the page, the meaning of the text, and the model training:

1. **OCR.** Each flyer is processed with OCR to extract the individual words together with their position on the page.
2. **LLM labeling.** An LLM assigns an entity label to each word, producing automatically labeled training data. We plan to run this via an academic cloud API key rather than a paid commercial account.

3. **Model training.** The labeled, structured output is then used to train a LayoutLM-style model that learns to perform the extraction itself.

Target corpus: up to several thousand offers, which is realistic given that the flyers are already collected automatically.

## Baseline Architecture

At a high level, the system follows three stages: OCR to obtain words and their positions, an LLM to label that text, and a trained LayoutLM-style model that learns to reproduce the extraction without the LLM. More concretely, we plan to implement:

1. **Layout-aware model.** We intend to use `microsoft/layoutlm-base`, a multilingual model with native German support that takes both the text and the position of each word into account. This follows the original LayoutLM approach (Xu et al., 2020).
2. **Subword alignment.** A module that maps our word-level labels onto the model’s subword tokens (labeling the first subword and masking the rest).
3. **Classification head.** A linear layer on top of the model that predicts the entity label for each token.
4. **Text-only baseline.** For comparison, a text-only German model (GBERT) with the same head but without position information.

## Work Plan

### Phase 1: Data Preparation

- Collect flyers and run OCR to get words and their positions.
- Set up the LLM labeling step and generate labeled training data.
- Bring the labels into a consistent BIO format and align them with the model’s subword tokens.

### Phase 2: Model Training

- Fine-tune the LayoutLM model on the labeled data.
- Train the text-only GBERT baseline for comparison.

### Phase 3: Evaluation

- Evaluate with entity-level Precision, Recall, and  $F_1$  (using `segeval`).
- Compare the trained model against the existing LLM black box.
- Look at where the model struggles (e.g. multi-product offers, OCR errors).

## Requirements

### Minimum

- A working end-to-end pipeline: flyer in, structured offers out, driven by our own trained model.
- Documented entity-level  $F_1$  for the trained model.

### Good

- A comparison between the layout-aware model and the text-only baseline, showing whether position information helps.
- A short error analysis of the most common failure cases.

### Excellent

- A comparison of the trained model against the LLM black box, including a discussion of the trade-offs (quality, cost, running locally vs. via API).

## Relation to the Course

The project aims for a complexity comparable to **Akbik et al. (2018)**, our course reference for learning-based sequence labeling: we train our own model on a self-built dataset, work with a layout-aware architecture, and compare it against a text-only baseline and the existing LLM system.

## References

- Akbik, A., Blythe, D., & Vollgraf, R. (2018). Contextual String Embeddings for Sequence Labeling. In *Proceedings of the 27th International Conference on Computational Linguistics* (pp. 1638–1649).
- Xu, Y., Li, M., Cui, B., Huang, S., Wei, F., & Zhou, M. (2020). LayoutLM: Pre-training of Text and Layout for Document Image Understanding. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* (pp. 1192–1200).
- (More papers to be added as the literature review continues.)*